Bioresource Technology 423 (2025) 132254

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

A machine-learning approach to optimize nutritional properties and organic wastes recycling efficiency conversed by black soldier fly *(Hermetia illucens)*

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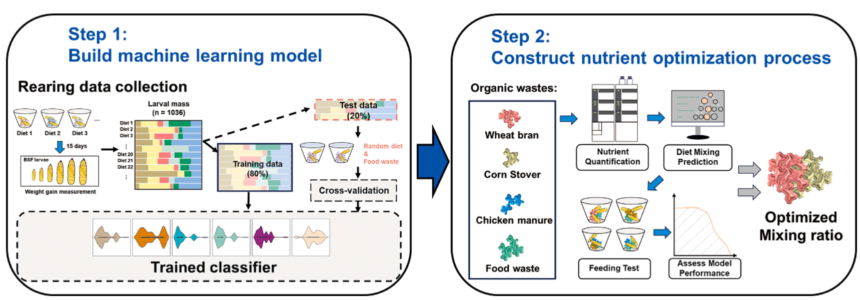
HIGHLIGHTS

- Diet protein significantly impacts the bioconversion rate of black soldier fly (BSF).
- The XGBoost model constructed using BSF rearing data could predict larvae end production.
- Optimal BSF nutrition was calculated from the 30,000 distinct random diets.
- XGBoost model-guided biowastes mixing strategy improves BSF bioconversion rate.

GRAPHICAL ABSTRACT



ABSTRACT

Suboptimal nutrition in organic waste limits the growth of black soldier fly (BSF) larvae, thereby reducing biowaste recycling efficiency. In this study, weight gain data from BSF larvae fed diets with distinct nutrient compositions were used to build a machine learning model. Among the algorithms tested, the XGBoost model demonstrated the best performance in predicting weight gain. The model identified protein as the most critical nutrient factor for larval biomass and was used to determine the optimal diet by calculating the highest weight gain from 30,000 randomly generated nutrient combinations. Supplementing the missing nutrients in organic waste according to the optimal diet improved the weight gain and feed conversion rate of BSF larvae. Feeding larvae a mixture of organic wastes, a cost-effective strategy to meet dietary nutrition requirements, resulted in significant increases in both the bioconversion rate (up to 9.7%) and mass reduction rate (up to 22.8%) of organic waste.

1. Introduction

The accumulation of organic wastes, including animal manure, agricultural residues, and food waste, poses critical environmental and societal challenges. This waste contributes to greenhouse gas emissions, contaminates water resources, and creates breeding grounds for pests

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https://doi.org/10.1016/j.biortech.2025.132254
Received 26 October 2024; Received in revised form 21 January 2025; Accepted 16 February 2025
Available online 17 February 2025

and pathogens, threatening both environmental sustainability and public health (Kharola et al. 2022). To reduce the environmental impacts of organic waste and maximise the recycling of potential nutrients (such as nitrogen and carbon) into valuable resources, traditional waste management strategies have been developed, including landfilling, incineration, and composting. However, these practices continue to face challenges related to energy efficiency, time consumption, and environmental impacts (Liu et al. 2022). Black soldier fly (BSF) (*Hermetia illucens*) farming has emerged as a promising solution for organic waste management. BSF larvae are highly efficient decomposers of organic matter and are capable of converting various waste streams into valuable biomass (van Huis and Gasco 2023). This biomass can be harvested as protein-rich feed for livestock and aquaculture, thus closing the loop in nutrient recycling and reducing reliance on conventional feed sources (Siddiqui et al. 2022). Additionally, the faeces produced by BSF larvae can be used as high-quality organic fertilisers, further enhancing the sustainability of agricultural practices.

One of the critical challenges in the BSF biowaste recycling industry is that organic waste cannot provide sufficient or balanced nutrition to BSF larvae, and the optimal nutrient conditions for BSF larvae have not been well studied. BSF larvae often feed on various readily available organic materials without considering whether their nutrient composition is suitable for BSF growth. Additionally, some biowastes, such as corn stover, are enriched with certain nutrients (including vitamin B) but are deficient in others, such as proteins, resulting in unbalanced nutritional conditions. This imbalance not only hinders the development of larvae but also causes inefficiencies in nutrient utilisation (Raksasat et al. 2020). Therefore, addressing these nutritional gaps is essential to improve BSF farming efficiency and biowaste recycling. To date, research on the nutritional requirements of BSF larvae has mainly focused on the protein-to-carbohydrate (P:C) or carbon-to-nitrogen (C:N) ratios in their diets because of the significant growth-promoting effects of these two macronutrients (Siddiqui et al. 2022). Larvae reared at specific P:C ratios have been shown to increase the bioconversion efficiency of organic wastes and enhance their end products (Jin et al. 2022; Rho and Lee 2022). However, the basic requirements of BSF larvae for other nutrients, such as fat, group B vitamins, and vitamin C, remain unclear. Whether these nutrient factors jointly affect the BSF bioconversion rate has yet to be determined.

Machine learning is a powerful tool that applies various algorithms to analyse data, learn from the features of items, and make accurate predictions based on the gathered information (Raschka et al. 2020). Machine learning techniques have been increasingly applied in agricultural and sustainable resource management research owing to their ability to model the complex relations between multiple features generated throughout the production process (Liakos et al. 2018). Traditional approaches for formulating BSF nutrient requirements rely primarily on trial-and-error experiments, which are resource-intensive and time-consuming. The complex interactions among nutrients and their cumulative effects further complicate the determination of an optimal diet for BSF larvae. Given these challenges, machine learning strategies offer a promising approach for optimising diet formulations by predicting how different nutrient combinations influence larval growth and development.

In this study, a machine learning-based approach was developed to calculate the optimal nutrient requirements of BSF larvae and provide guidance for optimising BSF diet nutrition (Fig. 1). Firstly, a series of weight gain data were collected from BSF larvae fed diets with distinct nutrient compositions. Several algorithms, including Decision Tree (DT), Random Forest (RF), and extreme gradient boosting (XGBoost), were applied to develop a predictive model using the collected data to predict the final larval mass based on diet nutrient supplementation. The XGBoost model exhibited the best performance. Secondly, totally 30,000 random diet [RD] combinations were simulated and identified the optimal nutrient composition for maximising larval biomass. Lastly, based on the machine learning model and the optimal diet nutrient

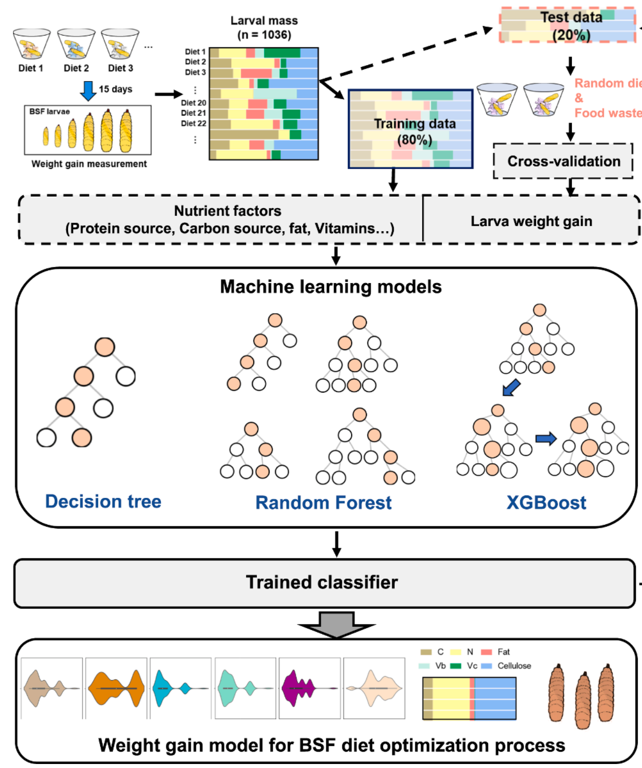


Fig. 1. Experimental design of this study. BSF larvae were initially reared on 25 diets with distinct nutrient supplementation for 10 d. The weight gain data were split to build machine learning models. After the best model was trained, random diet and organic waste diet rearing data were used for cross-validation. Finally, the validated classifier was used to predict BSF weight gain and conversion efficiency based on the diet's nutritional composition. BSF, black soldier fly. XGBoost, extreme gradient boosting; CART, Classification and Regression Trees.

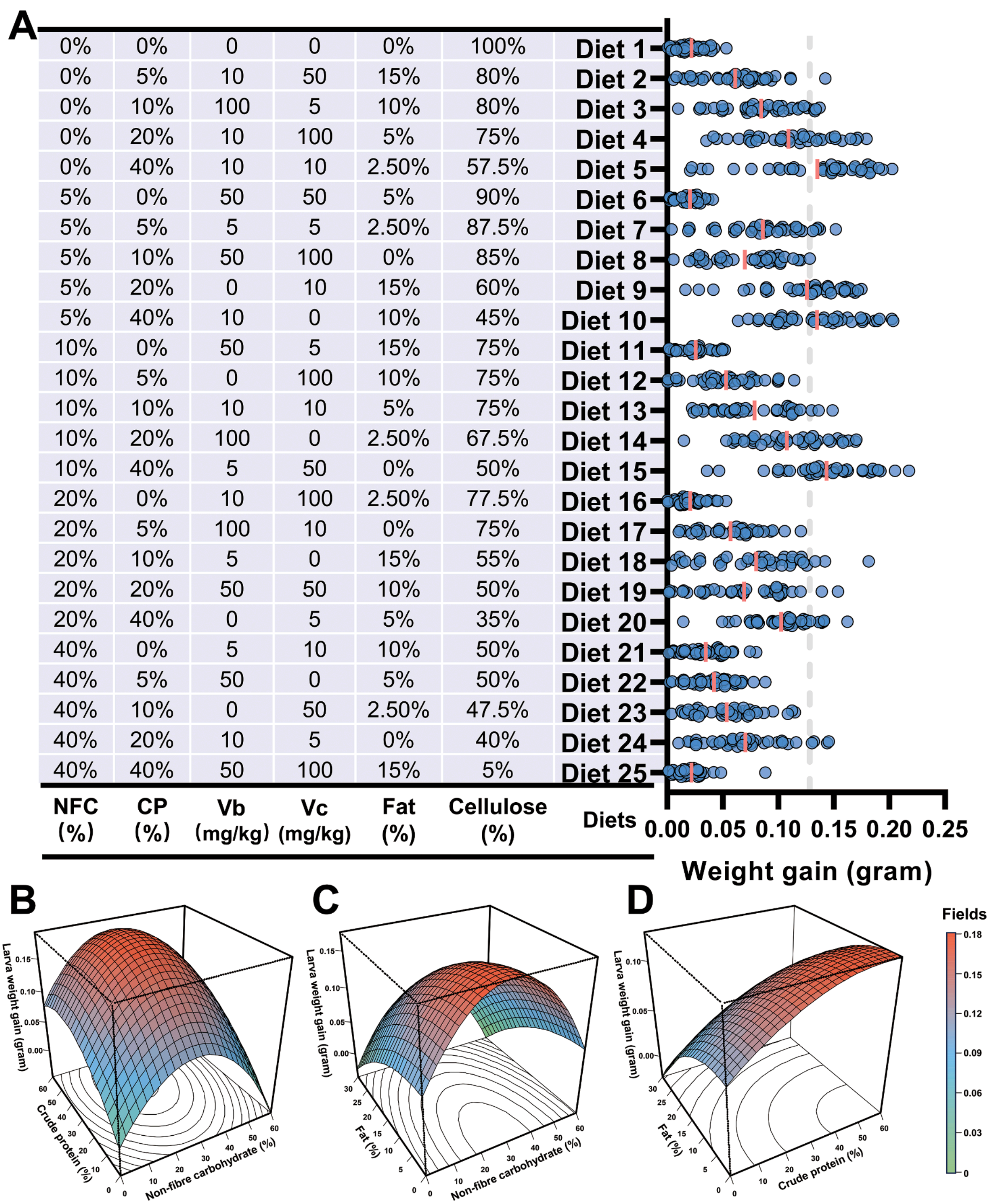
composition data, the BSF diet nutrient optimisation strategy has been investigated. The result suggests that proper feed additives for BSF larvae or calculates an optimal mixing ratio of different organic wastes can relieve the nutrient deficiency in organic wastes. This approach enables a cost-effective farming process to address the nutritional deficiencies of certain biowaste materials. Using the XGBoost model, BSF fed on optimised diets resulted in higher larval production (up to a 28 % increase in larval biomass after appropriate mixing of organic wastes) and significantly increased the conversion rate of organic wastes (up to 9.7 %). By incorporating machine learning methods, this study paves the way for the development of precise nutrient formulations in BSF farming and enhances larval end production and biowaste recycling efficiency in the organic waste management industry.

2. Materials and methods

2.1. Data Collection

2.1.1. Black soldier fly rearing data

The BSF eggs used in this study were collected from a laboratory-maintained population. After hatching in an incubator at 28 ± 1°C, 70 ± 5 % RH, and a photoperiod of 8:16 (light:dark), the larvae were reared under the same conditions and fed on the Gainesville diet for 5 d (Hogsette 1992). Subsequently, 2nd instar BSF larvae were screened using 2 mm and 2.8 mm mesh to collect larvae with similar body weights. These larvae were reared on artificial diets. To collect weight gain and its correlated diet nutrient composition data, larvae of similar sizes were weighed and reared on distinct artificial diets (N = 30 for each diet). A total of 25 distinct diets were generated using Taguchi orthogonal arrays (Fig. 2A; see Supplementary Materials) and consisted of casein acid hydrolysate (vitamin-free, as the crude fat), sucrose (non-fibre carbohydrate [NFC]), glucose (NFC, 1:1 with sucrose), soybean oil (fat), group B vitamin mixture, vitamin C, CMC-Na (inert carrier, 1:4 with cellulose powder, up to 100 % dry weight of diet), and cellulose powder (inert carrier). The moisture content of each diet was maintained at 60 %. Each ingredient was initially quantified and then prepared based on the formula according to its actual nutrient components (see Supplementary Materials). For each group, the larvae were maintained in a 10 × 10 × 5 cm plastic container with 10 g of artificial diet. The containers were covered with fine-meshed cloth. Similarly, larvae used for the nutrient gradient test (Fig. S2, see Supplementary Materials) were reared under the same conditions (N = 20 for each group). All larvae were reared for 10 d, collected, and weighed for further analysis.



NFC (%)	CP (%)	Vb (mg/kg)	Vc (mg/kg)	Fat (%)	Cellulose (%)	Diets
0%	0%	0	0	0%	100%	Diet 1
0%	5%	10	50	15%	80%	Diet 2
0%	10%	100	5	10%	80%	Diet 3
0%	20%	10	100	5%	75%	Diet 4
0%	40%	10	10	2.50%	57.5%	Diet 5
5%	0%	50	50	5%	90%	Diet 6
5%	5%	5	5	2.50%	87.5%	Diet 7
5%	10%	50	100	0%	85%	Diet 8
5%	20%	0	10	15%	60%	Diet 9
5%	40%	10	0	10%	45%	Diet 10
10%	0%	50	5	15%	75%	Diet 11
10%	5%	0	100	10%	75%	Diet 12
10%	10%	10	10	5%	75%	Diet 13
10%	20%	100	0	2.50%	67.5%	Diet 14
10%	40%	5	50	0%	50%	Diet 15
20%	0%	10	100	2.50%	77.5%	Diet 16
20%	5%	100	10	0%	75%	Diet 17
20%	10%	5	0	15%	55%	Diet 18
20%	20%	50	50	10%	50%	Diet 19
20%	40%	0	5	5%	35%	Diet 20
40%	0%	5	10	10%	50%	Diet 21
40%	5%	50	0	5%	50%	Diet 22
40%	10%	0	50	2.50%	47.5%	Diet 23
40%	20%	10	5	0%	40%	Diet 24
40%	40%	50	100	15%	5%	Diet 25

Fig. 2. Relation between diet nutrition and BSF larval biomass. (A) Nutrient composition of different diets (left) and corresponding weight gain data of BSF larvae (right). Larvae fed Diets 5, 10, and 15 showed higher weight gain compared with that of those fed other diets. The dashed line represents the average weight gain of BSF larvae fed on food waste. (B–D) Response surface plots showing the effect of nutrient interactions on larval mass: (B) Effect of dietary non-fibre carbohydrates (NFC) and crude protein (CP) on larval biomass, (C) Effect of dietary NFC and fat (ether extract) on larval biomass, (D) Effect of dietary CP and fat on larval biomass. BSF, black soldier fly; VC, vitamin C; VB, Vitamin B.

2.1.2. Quantification of Macronutrients in black soldier fly larva and biowaste Materials

Nutrient factors including crude protein (CP), crude fat (EE), neutral detergent fibre (NDF), ash content (ASH), and non-fibre carbohydrates (NFC) were also measured in artificial diet ingredients and organic waste to assess the quality of BSF diet. Crude protein content was calculated according to the nitrogen content using the Kjeldahl method, and the resulting value was further multiplied by 4.76 as a conversion factor. The crude fat content of BSF larvae was measured by AOAC Soxhlet extraction method 920.39 employing diethyl ether to obtain fat from dried BSF larva samples. Neutral detergent fibre (NDF) contents in organic wastes were initially treated by heat-resistance amylase and then calculated according to Van Soest's method (Van Soest et al. 1991). The ash content was measured by burning samples in a muffle furnace at 600℃. The remaining ash was weighted for the final ash content. As for non-fibre carbohydrates, this nutrient factor was calculated by Eq. (1) (Ferreira and Mertens 2005):

$$NFC = 100\% - (CP\% + NDF\% + EE\% + Ash\%) \quad (1)$$

2.1.3. Quantification of vitamins in black soldier fly diets

Liquid Chromatograph Mass Spectrometer (LC-MS) analysis was performed to determine the vitamin content of artificial diet ingredients (Casein, soybean oil) and organic wastes (wheat bran, corn stover, chicken manure, and food waste). This system relies on a chromatographic column for chemical separation and mass spectrometry for ion identification and quantification. To extract vitamins, 5 mg of each sample was weighted and transferred into a 1.5 mL tube. A total of 300 μL of extraction buffer (100 μM DTT in H_2O) was added into the tube and subsequently homogenized by a hand-held homogenizer (Fatima et al. 2019). The homogenized samples were incubated in a water bath at 90 °C for 10 min and then centrifuged at 4,000 g for 5 min. The pellet was discarded, and the supernatants were filtered by a 0.22 μm filter (Millipore). The filtered samples were loaded into LC-MS (2 μL).

Vitamin B and C in samples were quantified by an Ultra-high Performance Liquid Chromatography with Quadrupole Time-of-Flight Mass Spectrometry (UPLC-QTOF-MS) system (Waters, USA) operating in negative mode and using an MS scan from m/z 100–1000. The ion spray voltage was set to 5.5 kV for positive mode (ESI +). The drying gas temperature was set at 400 °C. LC separation of target components was performed using an ACQUITY UPLC BEH C18 column (100 mm × 2.1 mm × 1.7 μm, Waters). The mobile phase A consisted of 25 mM ammonium hydroxide and 25 mM ammonium acetate in water, and the mobile phase B consisted of methanol. The LC gradient program was set as follows: 99 % A for 0.5 min; decrease to 92 % A within 2 min, then decrease to 50 % A for 3 min; 5–8 min decrease to 10 % A; 8–9 min maintained at 10 %; finally increased to 95 % for 3 min and maintained for 1 min. The temperature of the sampler was kept at 4 °C (Fatima et al. 2019). The LC-MS result data was further analysed by MassLynx software version 4.1. Totally 100 μM of bromophenol blue sodium salt was added into each sample as the internal standard to calculate the final concentration of vitamins in BSF diets.

2.2. Data preprocessing

Before building the machine learning models, a data preprocessing step was performed to ensure the consistency and structure of the input data, which made it suitable for further machine learning analysis. Considering that the BSF weight gain data were collected from larvae that survived the 10-d feeding period, no missing weight gain or nutrient values were identified. Moreover, duplicate data (repeated entries) were removed from the dataset (Muinde et al. 2023). After the data cleaning step, data set was standardised using the StandardScaler function implemented in the sklearn preprocessing library (Adjuik et al. 2024).

2.3. Exploratory data analytics (EDA)

The gathered standardized data were initially subjected to exploratory data analytics. Distribution of each feature was checked and the kernel density estimates for the weight gain data was calculated by the 'kdeplot' function of seaborn library in Python platform. Finally, correlation analysis between larva weight gain data and nutrient contents in diets was calculated by Python and R platform ('cor()' function) respectively using Pearson correlation and visualized by seaborn heatmap.

2.4. Model building process

All features (nutrient factors and larval biomass data; Table 1) were used for generation of machine learning models. Totally 80 % of the pre-processed weight gain data were used as training samples, whereas the remaining 20 % were used as test data to validate the models. Three algorithms (DT, RF, and XGBoost) were applied to generate a nutrient-based weight gain model. All models were developed using the scikit-learn package in Python.

2.4.1. Decision tree (DT)

Decision Tree is a widely used machine learning strategy and it operate by recursively partitioning the input feature space into distinct regions based on specific criteria, forming a tree-like structure (Loh 2011). Decision tree can handle both numerical and categorical data, making them versatile for various applications. In this study, Decision Tree Regression—a specific type of DT used for predicting continuous features (including nutritional contents in the BSF diet and weight gain data)—was employed (Geurts et al. 2009). The DT used for model construction was based on the Classification and Regression Trees algorithm (Loh 2011). This algorithm recursively splits the feature space into binary partitions, where each node corresponds to a decision based on a feature, and each leaf node corresponds to the predicted output value. In this study, DT model was built by DecisionTreeRegressor() function of scikit-learn package.

2.4.2. Random Forest (RF)

In contrast to the DT method, the RF model is an ensemble learning strategy that creates multiple DTs during training (Greener et al. 2022). Each tree was built using a random subset of the training data and a random subset of features at each split. The final prediction was made by aggregating the predictions of all individual trees. Although RF offers strong predictive performance, its interpretability is lower than that of a single DT. However, it provides useful metrics, such as feature importance, which indicate the contribution of each feature to the model's predictions. In this study, the nutrient factors of the BSF diet were used to build distinct trees to determine the final weight gain of BSF larvae (by function RandomForestRegressor() in scikit-learn package). The

Table 1
Variables used for generating machine learning models.

Variables*	Description
NFC (%)	Non-fibre carbohydrate content (w/w in dry diet) measured in diets. It is the carbon source for the development of black soldier fly (BSF) larva.
CP (%)	Crude protein content (w/w in dry diet) measured in diets. It is the nitrogen source of BSF larva.
VB (mg/kg)	Total amount of group B vitamins (mg/kg in dry diet) measured in diets. They are micronutrients that are critical for BSF larva.
VC (mg/kg)	Vitamin C content (mg/kg in dry diet) quantified in diets.
Fat (%)	Fat content measured (w/w in dry diet) in diets.
Cellulose (%)	Mixture of CMC-Na and cellulose (1:4) added in the artificial diet as the inert carrier.
Weight gain (g)	Biomass of fresh BSF larva after a 10-d feeding period.

*NFC, non-fibre carbohydrates; CP, crude protein; VB, vitamin B; VC, vitamin C.

randomness of these trees reduced variability among individual trees, efficiently minimising overfitting and increasing overall performance.

2.4.3. *eXtreme gradient boosting (XGBoost)*

XGBoost is a tree-based algorithm that uses a sequential ensemble of tree models (Mayr et al. 2014). Unlike RF, which builds trees independently, XGBoost constructs each tree to correct errors in the previous one. The model was trained using both first- and second-order gradient optimisation, along with regularisation techniques to prevent overfitting (Kirk et al. 2022). Using this strategy, each input feature (different concentrations of nutrient factors) was assigned a weight and predicted using a DT. Subsequently, the weights of the features that produced incorrect predictions were upregulated and sent to the next DT for further prediction. These individual classifiers were combined to create a novel model with high performance. Xgboost models were built by function 'XGBRegressor()'. SHapley Additive exPlanations (SHAP) analysis of distinct features in the XGBoost model was performed using the R package xgboost and SHAPforxgboost (ver. 0.1.1, R script used in this study have been submitted to https://github.com/Bosheng13/BSF-weight_gain_model/blob/main/BSF_Weightgain_Rscript.ipynb).

2.5. *Performance evaluation*

Machine learning models were initially generated using default parameters. To evaluate performance, several indices were applied, including the coefficient of determination (R^2) root mean square error (RMSE), and mean absolute error (MAE) (Kirk et al. 2022). The R^2 Value (Eq. (2)) measures how well the regression model fits the data:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(x_i - \widehat{x}_i)^2}{\sum_{i=1}^{n}(x_i - \overline{x}_i)^2} \tag{2}$$

In this equation, x_i is the actual weight gain value of BSF larva, $\widehat{x}_i$ is the predicted weight gain value, and $\overline{x}_i$ is the mean value of actual weight gain data.

The RMSE indicate the square root of the mean squared error (MSE) of model, giving an error in the same unit as the target variable. This index can be calculated by Eq. (3):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \widehat{x}_i)^2} \tag{3}$$

The last index MAE measures the average absolute difference between actual and predicted values, which can be calculated by Eq. (4):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - \widehat{x}_i| \tag{4}$$

R^2 value should range from zero to one, and RMSE and MAE value should be non-negative values. The higher R^2 value suggests that the model explains most of the variance in BSF larval weight gain using the provided nutrient composition, which indicate a higher performance of the model (Li 2017). Meanwhile, a lower level of RMSE and MAE value suggests larval weight gain values predicted by the generated model is close to the actual measured weight gain. This suggests that the nutrient composition data used in the model can effectively captures the factors influencing larval growth.

2.6. *Performance optimisation*

To enhance models' performance, both the grid search method and Gray Wolf Optimiser (GWO) were used to determine the best combination of parameters. Grid Search is an exhaustive search algorithm used to tune parameters by evaluating all possible combinations within a predefined range. It systematically explores a grid of parameter values and selects the combination that results in the best model performance based on a chosen evaluation metric. Differently, GWO is more

Table 2
Parameters and performance of models built by different algorithms.

	Major parameters	Default model				Grid search optimised model				Gray wolf optimiser optimised model			
		Parameter values	R^2	RMSE	MAE	Optimised parameters	R^2	RMSE	MAE	Optimised Parameters	R^2	RMSE	MAE
Decision Tree	**max_depth** (Maximum depth of the tree)	None	0.8083	0.0224	0.0181	12	**0.8083***	**0.0224***	**0.0181***	13	0.8073	0.0225	0.0181
	max_features (Number of features to consider for best split)	None				5				1			
	min_samples_split (Minimum number of samples required to split a node.)	2				2				2			
	min_samples_leaf (Minimum number of samples required at a leaf node.)	1				5				6			
Random Forest	**n_estimators** (The number of trees in the forest.)	100	0.8084	0.0224	0.0181	50	**0.8087***	**0.0224***	**0.018***	54	0.8086	0.0224	0.018
	max_depth (The maximum depth of the tree.)	None				20				28			
	min_samples_split (The minimum number of samples required to split an internal node)	2				10				2			
	min_samples_leaf (The minimum number of samples required to be at a leaf node.)	1				1				2			
	max_features (The number of features to consider when looking for the best split.)	auto				5				1			
XGBoost	**n_estimators** (The number of boosting rounds (trees) to train)	100	0.8082	0.0224	0.0181	510	0.8084	0.0224	0.0181	2566	**0.8093****	**0.0223****	**0.018****
	learning_rate (Step size shrinking to prevent overfitting.)	0.1				0.3				0.583286224			
	max_depth (Maximum depth of each tree)	6				3				3			

R^2, coefficient of determination; RMSE, root mean square error; MAE, mean absolute error;
*, best performance within same type of machine learning models.
**, best performance within all built machine learning models.

computationally efficient algorithm based on the hunting behaviour of grey wolves. It mimics the leadership hierarchy and cooperative hunting strategies of wolf packs to search for optimal solutions. In this algorithm, the grey wolves encircle during their hunting process, and their encircling behaviour can be mathematically modelled by these equations:

$$\vec{D} = \left| \vec{C} \bullet \vec{X}_p(t) - \vec{X}(t) \right| \tag{5}$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \bullet \vec{D} \tag{6}$$

In Eq. (5), t indicates the current iteration. $\vec{D}$ represents the distance from the current wolf's position to the best wolf's position vector $\vec{X}_p(t)$, and $\vec{X}(t)$ is the current position vector of the grey wolf. $\vec{C}$ is a coefficient vector that calculated by Eq. (7):

$$\vec{C} = 2 \bullet \vec{r}_2 \tag{7}$$

Eq. (6) shows the nest position vector (next iteration) of the grey wolf, $\vec{A}$ is also a coefficient vector that can be calculated by Eq. (8):

$$\vec{A} = 2\vec{a} \bullet \vec{r}_1 - \vec{a} \tag{8}$$

The value of $\vec{a}$ are linearly decreased from 2 to 0 during the iterations and both the vector $\vec{r}_1$ and $\vec{r}_2$ are vectors randomly changed from 0 to 1.

In this study, machine learning parameters such as Maximum depth of each tree (max_depth), Step size shrinking to prevent overfitting (learning_rate), and the number of boosting rounds (trees) to train (n_estimators) can be used to define the position of each wolf in the search space. For example, the position of a grey wolf can be represented as: $\vec{X}$ = [max_depth, learning_rate, n_estimators]. As these parameters change during the optimisation process, the position of the grey wolf is adjusted accordingly. When the wolves converge to the position of their prey (the best wolf's position vector $\vec{X}_p(t)$), each element of this position vector could be the optimised values of the machine learning parameters (Mirjalili et al. 2014). GWO dynamically updates solutions by balancing exploration and exploitation, allowing it to efficiently search complex and high-dimensional parameter spaces. In both optimisation methods, the parameters that achieved the highest R^2 and the lowest RMSE and MAE values for the target model were selected as the final input parameters (Table 2, Python script: https://github.com/Bosheng13/BSF-weight_gain_model/blob/main/BSF_Weightgain_Python.ipynb).

2.7. Conversion efficiency of mixed-diet feeding larvae

After obtaining the optimal BSF diet using the XGBoost model, we next aimed to determine whether adding the missing nutrients to organic wastes (through feed additives) could enhance larval biomass and biowaste conversion efficiency. The feed additives used in this study were similar to the ingredients in the artificial diets (see section 2.1 and supplementary materials). BSF larvae were reared using the method described in 2.1.

The mass reduction rate of organic wastes treated by BSF larva was calculated by the formula (ur Rehman et al. 2017):

$$Mass\ \ reduction = M1 - \frac{M2}{M1} \times 100 \tag{9}$$

Where M1 is the total mass of the organic waste provided to the BSF larva, M2 is the total mass of the organic waste treated by the BSF larva.

Feed conversion efficiency indicate the digestive ability of BSF larvae feeding on distinct materials. This index can be calculated as (Bosch et al. 2019):

$$\text{Feed conversion efficiency} = \frac{Total\ \ mass\ \ of\ \ ingested\ \ food\ \ (gram)}{Total\ \ mass\ \ of\ \ weight\ \ gain\ \ (gram)} \times 100 \tag{10}$$

To ensure that BSF larvae consume both components of the mixed diet (e.g., the mixture of wheat bran and crop straw) and do not preferentially feed on the more nutritious ingredient (wheat bran), implemented controlled feeding conditions were used in this study. Briefly, the total amount of diet supplied to larva was limited (1.5 kg). A total of 1,000, 2,000, 5,000, and 10,000 eggs (10,000 eggs per gram) were introduced into 1.5 kg diet (single organic waste material) respectively to determine the best larval density that consumes all diet components and meanwhile exhibited a high total biomass. Larvae were reared in a 40 x 30 x 15 cm plastic container for each group. The top of the box was covered by a black cloth. All containers were kept in the incubator with a light/dark cycle of 8:16, the temperature at 28 ± 1℃, and 70 ± 5 % RH. The optimal diet (Fig. 4A) was used as the Control group. Larvae (eggs) and diets were kept in the incubator at 28 ± 1°C, 70 ± 5 %RH for 15 days. Afterward, all larvae were separated from the diet, and their total mass was weighted. Groups with the highest total larvae mass were selected for further experiments that calculate the bioconversion rate of mixed diets using the formula:

$$Bioconversion\ \ rate = \frac{Total\ \ mass\ \ of\ \ BSF\ \ larave\ \ (gram)}{diet\ \ used\ \ for\ \ BSF\ \ rearing\ \ (gram)} \times 100 \tag{11}$$

2.8. Data analysis

Variation analysis of data was performed by Mann-Whitney *U* test and pairwise Wilcox test using 'wilcox.test()' and pairwise_wilcox_test ()' in R respectively. Response surface analysis of diet nutrients was performed by "rsm" packages implemented in R (R scripts was submitted to GitHub: https://github.com/Bosheng13/BSF-weight_gain_model/blob/main/BSF_Weightgain_Rscript.ipynb).

3. Results and Discussion

3.1. Basic nutrient requirements of BSF larvae

Weight gain data for 1306 BSF larvae were obtained from 25 distinct dietary groups. After a 10-day feeding period, BSF larvae in all test groups showed a high survival rate (> 80 %, see Supplementary Materials). Larvae reared on Diets 5, 10, and 15 showed the highest weight gain (Fig. 2A). The crude fat and protein content in BSF larvae showed similar results (see Supplementary Materials). Notably, all three diets contained a high amount of CP, whereas the NFC, vitamin C, and fat content in these diets were relatively low. This suggests that CP is one of the most critical nutrient factors required by BSF larvae during development. As the vitamin-free casamino acids used in this study contained low amounts of carbohydrates and fat (see Supplementary Materials), BSF larvae likely met their basic requirements for these nutrients from the high concentration of CP provided. Based on these data, response surface analysis was performed to explore the optimal combination of nutrient factors. The results indicated that dietary NFC and CP in a combination of 25 % and 40 % of the dry weight, respectively, exhibited the highest weight gain (Fig. 2B). The best combination of NFC and fat was ~ 27 % NFC with 2.5 % fat, which further enhanced the weight gain of the BSF larvae (Fig. 2C). When CP and fat were combined, the response surface was not curved, and high CP content (>40 %) was observed to induce greater weight gain in BSF larvae (Fig. 2D). Interestingly, when all nutrient factors were supplied at high concentrations, BSF larvae exhibited a very slow growth rate, similar to that observed with the diet lacking additional nutrients (Diet 1, which contained only cellulose, Fig. 2A). These findings suggest that over supplementation of nutrients in the BSF diet can disrupt nutrient conversion in larvae. Thus, preparing an appropriate nutrient composition in the diet is both a resource-saving and high-yielding strategy for BSF farming and the organic recycling industry.

Compared with landfill disposal and anaerobic digestion strategies, organic waste recycling through BSF farming is time-efficient, and its products exhibit low odour emissions and strong economic feasibility (including high-quality insect protein and fat biomass) (Liu et al. 2022; Papa, Scaglia et al. 2022). BSF larvae can utilise various types of organic waste, including animal manure, food waste, vegetables, and fruit waste (Siddiqui et al. 2022), highlighting their high adaptability and waste conversion efficiency. Despite studies comparing larval yield when fed different materials and examining the corresponding nutritional quality of BSF larvae (Barragan-Fonseca et al. 2017), the basic nutritional requirements of this insect species remain unclear. Most nutritional investigations of BSF larvae have focused on the P:C or carbon-to-nitrogen ratios in their diets, yielding divergent results, including P17:C55 (Barragan-Fonseca et al. 2019), *P*1:*C*2 (Barragan-Fonseca et al. 2021; Eggink et al. 2023), and C18:N1 (Jin et al. 2022). This divergence arises from differences in methods used to calculate CP and carbohydrate contents. Considering that some nitrogen source in diet (e.g., ammonia nitrogen, nitrate, and urea) might not directly utilized by BSF larvae, it's better to measure protein content to estimate the nutrition quality of BSF diet. Similarly, carbohydrates such as cellulose and starch (Fig. S3E, see Supplementary Materials) cannot be easily digested should not be included for measuring diet nutrient quality. BSF larvae fed sucrose-supplemented diets achieve higher survival rates and dry waste reduction rates compared with those of larvae fed starch-supplemented diets (Cohn et al. 2022). Accordingly, starch supplementation in the BSF diet (Barragan-Fonseca et al. 2021) may lower the P:C ratio of the optimal BSF diet. Therefore, parameters such as reducing sugar or non-fibre carbohydrate can reflect the real nutrient requirement of BSF larva. In this study, sucrose and starch incorporated a 1:1 mixture into the artificial diet, similar to the method used previously (Cammack and Tomberlin 2017). The results of this study support previous findings, demonstrating that a relatively balanced P:C ratio (33 %:27 % in this study) enables better growth in BSF larvae. Additionally, the dipteran relative of BSF, *Drosophila melanogaster*, also exhibited high larval mass when fed a balanced or slightly protein-biased diet (Lee and Jang 2014). Similar P:C ratios have been found to promote the development of lepidopteran and coleopteran species (Rho and Lee 2022).

3.2. Diminishing returns in Carbohydrate, Fat, and vitamin gain in BSF larvae

Because the nutrient content in the 25 diets might not fully cover the optimal concentrations required for BSF development (including the concentration of CP; Fig. 2D), a broader range of nutrient concentrations was applied to observe the weight gain of larvae. The diet with the highest weight gain (Diet 15; see Supplementary Materials) was selected as the baseline nutrient composition, and a concentration gradient of each nutrient factor (NFC, CP, ether extract, vitamin B, and vitamin C) was prepared to feed the BSF larvae. Except for CP, a diminishing return effect was observed as the concentrations of NFC, ether extract, and vitamins B and C in the diet increased (Fig. S2, see Supplementary Materials). The weight gain of BSF larvae was the highest when 10 % NFC was supplied (Fig. S2A, see Supplementary Materials). Similarly, adding 5 % fat to the diet resulted in the highest weight gain, whereas increasing dietary fat content led to low larval biomass, especially when the fat content reached 30 % (Fig. S2B, see Supplementary Materials). Similar trends were observed for vitamins B and C, with optimum concentrations of 10 mg/kg for the vitamin B mix and 100 mg/kg for vitamin C in the BSF diet (Fig. 3D and E). These findings highlight that a diminishing return effect for NFC, fat, and vitamins is present in BSF larvae, and this effect should be routinely considered when feeding larvae a nutritious diet (including food waste) to reduce costs and improve energy recycling efficiency. Remarkably, CP content exceeding 40 % did not result in further increases in larval biomass (Fig. S2B, see Supplementary Materials). Beyond determining the optimum nutrient requirements for BSF larvae, interactions between distinct nutrient factors and BSF larval weight were assessed. Spearman's correlation analysis revealed that only CP content was positively correlated with larval mass (R2 = 0.53; Fig. S2F, see Supplementary Materials), whereas other nutrient factors exhibited negative correlations ($R^2 < 0$) with larval weight gain because of the diminishing returns observed in the larvae. No significant correlations were found among the nutrient factors, apart from a strong negative correlation between cellulose and other nutrients caused by the fixed dry matter content in the BSF diet, where an increase in cellulose reduced the concentrations of other nutrients.

To explore the potential of deriving nutrient factors from diverse organic wastes, the impact of CP, NFC, and fat from distinct food materials on the weight gain of BSF larvae was tested. Larvae fed animal-origin casein showed no difference in growth compared to those fed plant-origin soy peptone (Fig. S2G, see Supplementary Materials), indicating that the protein source does not specifically affect BSF larval growth. Conversely, different types of carbohydrates and fats had a marked influence on larval weight gain. Sucrose in the diet significantly increased weight gain compared to starch-fed larvae (Fig. S2H, see Supplementary Materials), suggesting that oligosaccharides or monosaccharides may have a higher digestibility rate than that of polysaccharides in BSF larvae. Similarly, larvae fed animal-origin grease showed better growth than that of those fed animal-origin oil (Fig. S2I, see Supplementary Materials). These findings suggest that organic waste from animal sources can enhance BSF larval production. Mixing animal-origin waste with other materials, such as corn straw, may improve waste utilisation and BSF production.

Notably, the nutritional composition of most organic wastes does not perfectly meet the dietary requirements of BSF larvae (Wang et al. 2020), as some nutrient elements may be deficient or over-supplemented. A scientific approach to mixing these organic wastes enables BSF larvae to efficiently convert valuable nutrients into high-quality larval biomass. However, over-supplementation of certain nutrients in some biowastes (including NFC content in food waste) leads to diminishing returns, adversely affecting the waste conversion rate. Several biological processes may cause this effect. Firstly, larvae have reached their maximum capacity to assimilate the oversupplied nutrient factors. Excess ingestion of these nutrients may not be metabolized and excreted without contributing to biomass gain. Second, excessive amounts of certain nutrients can interfere with the absorption and utilization of others. For example, a high-protein diet may inhibit the metabolism of carbohydrates (Morrison and Laeger 2015). Lastly, over-supplementation of some nutrient factors, such as vitamins and minerals (e.g., calcium and copper), can lead to insect toxicity (Rivera-Pérez et al. 2017). To avoid diminishing returns in the BSF farming process, a precise nutrient requirement of this insect and a reliable method to optimise the nutrient composition of the BSF diet is urgently needed. The machine learning model can serve as a powerful decision-support tool and offer important parameters for improving the recycling efficiency of organic waste in the BSF farming industry.

3.3. Xgboost algorithm showed the best performance for the Nutrient-Based weight gain model

The predicted and true values of the BSF weight gain data at various training sizes across different algorithms (Fig. 3A, C, and E) were compared. The performance of the models generated by each algorithm was also evaluated (Fig. 3B, D, and F). Both algorithms demonstrated relatively high performance even when the training size was less than 100 (Fig. 3A, C, and E), and increasing the training size did not significantly affect the performance of weight gain prediction. A training size of 800 (approximately 80 % of the input data) was used to generate the

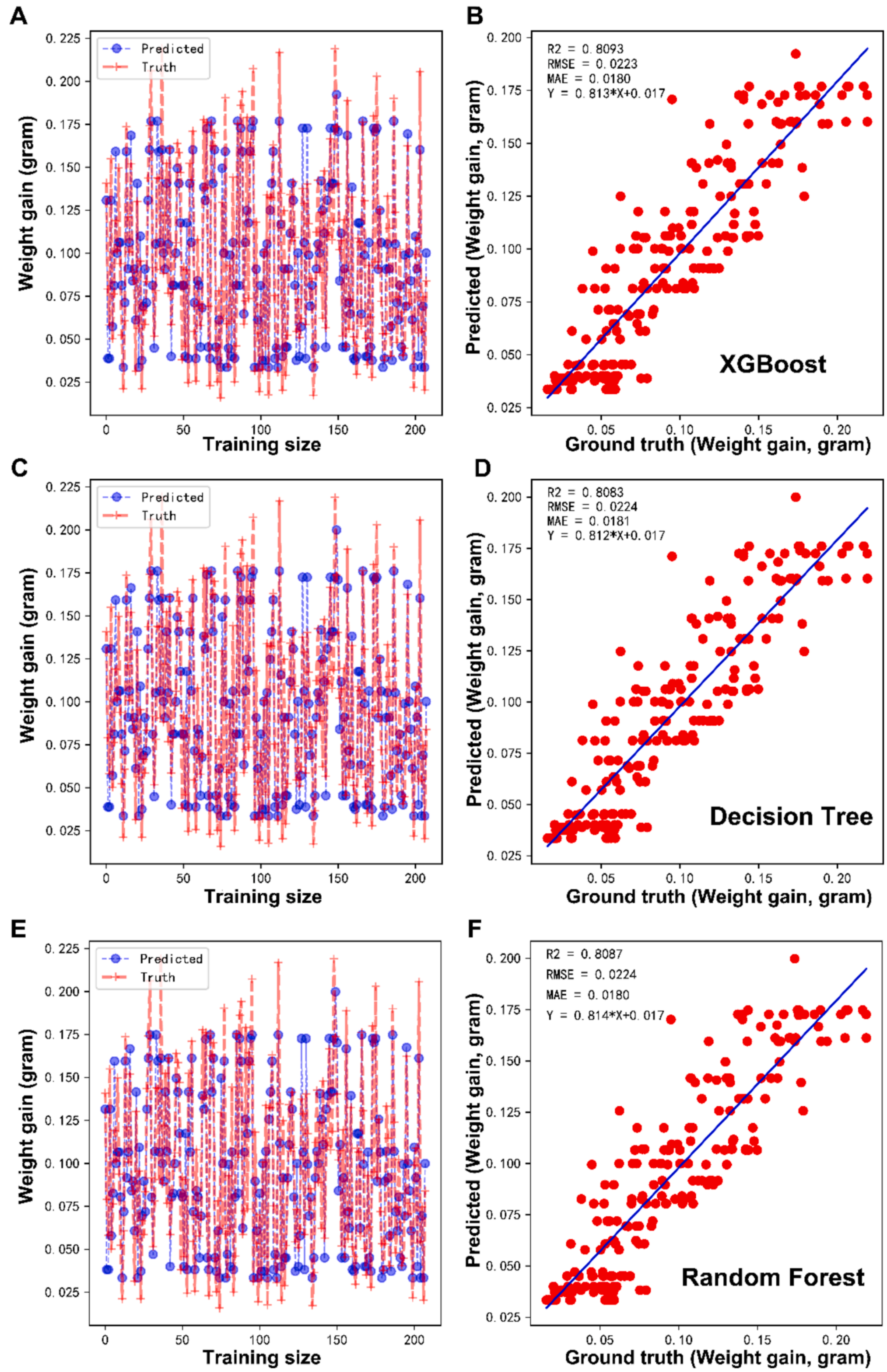


Fig. 3. Model performance comparison among XGBoost, Decision Tree, and Random Forest algorithms. (A, C, and E) Comparison of predicted (blue dots) and true weight gain values (red crosses) as the training size increases for XGBoost (A), Decision Tree (C), and Random Forest (E). (B, D, and F) Scatter plots of the training set showing predicted and actual weight gain values calculated by XGBoost (B), Decision Tree (D), and Random Forest (F). Indices used to describe the fitness of models, including root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2), are visualised. XGBoost, extreme gradient boosting; CART, Classification and Regression Trees.

final model. Performance of the three different machine learning strategies indicated that the GWO optimised XGBoost model exhibited superior predictive performance and relatively stable prediction ability, achieving R^2 values of 0.8094 and RMSE = 0.0223 (Table 2, Fig. 3B). Among the other algorithms, the Grid search optimised model showed the highest performance among RF models (R^2 = 0.803, RMSE = 0.224; Table 2, Fig. 3D) than the DT method (Grid search optimised model exhibited the highest performance, R^2 = 0.8088, RMSE = 0.0224; Table 2, Fig. 3F). IT should be noted that the RMSE and MSE values of all models showed no significant difference among each other. Considering the overall performance, the XGBoost algorithm was selected as the final strategy to generate a nutrient-based weight gain model for BSF larvae.

The ever-increasing demand for breeding products, combined with the growing amount of information collected through intelligent

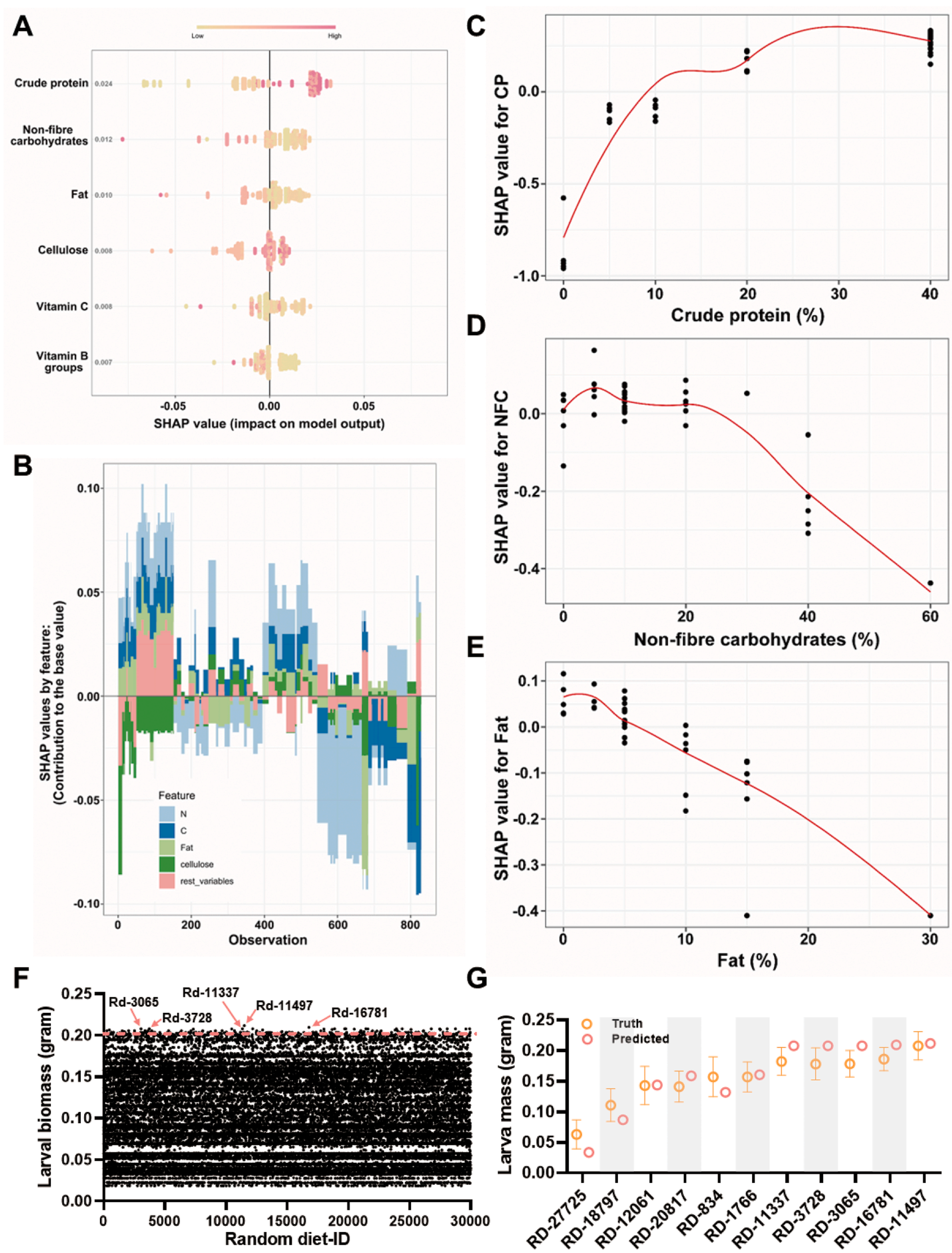


Fig. 4. Feature importance of nutrient factors calculated by the XGBoost model. (A) SHAP values indicating the importance of nutrient factors in the BSF diet. (B) SHAP interaction plot showing the contribution of each nutrient factor to each observation. (C–E) SHAP dependence plots for crude protein (C), non-fibre carbohydrates (D), and fat (E), with scatter plots and red regression lines representing the mean values. (F) Larval biomass data from 30,000 randomly generated diets predicted by the XGBoost model, with the top five diets marked by arrows. (G) Comparison of predicted (XGBoost model) and actual larval biomass (feeding data) for the five diets with the highest biomass among all 30,000 diets. NFC, non-fibre carbohydrate; CP, crude protein; SHAP, SHapley Additive exPlanations; XGBoost, extreme gradient boosting.

devices, such as artificial intelligence, has become a critical factor in achieving breeding goals for crops, livestock, and the recycling of residues and excreta (Guo et al. 2021; Hayes et al. 2023; Silveira et al. 2023). Machine learning methods perform well in food processing and livestock nutrition, promoting the development of precise nutrition (Durand et al. 2023; Menichetti, et al. 2023). However, artificial intelligence has been underutilised in BSF farming. Only Muinde et al. (2023) and Arshad et al. (2023) have reported that machine learning methods can help optimise rearing conditions to improve the waste conversion rate (Arshad, et al. 2023; Muinde et al. 2023). In this study, an XGBoost model was developed to uncover the relation between the nutrient requirements of BSF larvae and their final larval biomass production. Various tools are available for modelling complex patterns in agricultural and industrial applications, including Linear Regression (LR), Support Vector Regression (SVR), K-Nearest Neighbours, and deep learning methods, such as multilayer perceptron, convolutional neural networks, and recurrent neural networks (Raschka et al. 2020). The data used in this study were continuous and featured multiple nutrient factors that could interact with each other. The XGBoost model built in this study provides a key step towards optimising the nutritional requirements of BSF larvae while enhancing the efficiency of organic waste bioconversion. Given the complexity and number of features in the collected data (six nutrient factors), machine learning was chosen over deep learning to construct the prediction model (Liu et al. 2023). Tree-based algorithms (DT, RF, and XGBoost) were employed, as they effectively prevent overfitting in the current dataset. The XGBoost algorithm treats nutrient factors independently, building rules based on the values of those features. This approach allows for better interpretability of the collected data (Stanik et al. 2019). Nonetheless, the BSF weight gain database used to construct the prediction model should continue to be enriched, and additional nutritional factors should be included in future models. It is important to note that the feeding data (diet nutrient composition and correlated weight gain) could be expanded through further investigation, with the number of features increasing as more nutrient elements are included. Consequently, the XGBoost model used in this study may not remain the optimal algorithm for predicting BSF larval weight gain. In such cases, deep learning methods, such as Multi-Layer Perceptron, could be better suited for handling more complex datasets and improve the performance of the weight gain model (Liakos et al. 2018).

3.4. Protein content in diet is the most critical nutrient factor for BSF

Once the optimal model was determined, the impact of different nutrient elements in the BSF diet was assessed using the constructed XGBoost model. SHAP values were used to interpret the outcomes of the XGBoost model and estimate the importance of each nutrient factor. The SHAP analysis revealed that CP content was the most critical factor influencing the weight gain of BSF larvae (Fig. 4A), showing the highest SHAP value among all nutrient factors. NFC content ranked as the second most important factor, followed by fat content. Vitamins, particularly B-group vitamins, had a relatively lower impact compared with that of other nutrients (Fig. 4A). This may be because of the presence of vitamins in other components of the diet, such as casein and plant oil. The SHAP value compositions for each observation was also estimated using a SHAP force plot (Fig. 4B). In most cases, CP accounted for more than 50 % of the total SHAP value, consistent with the SHAP values calculated for each nutrient factor. Further analysis of SHAP values against changing nutrient contents showed that the SHAP value for CP increased as CP concentration increased (Fig. 4C–E), indicating the growing importance of CP with dietary enrichment. This finding aligns with results from feeding tests with gradient CP concentrations. Conversely, SHAP values for NFC and fat decreased with increasing levels of these nutrients (Fig. 4D and E), suggesting that higher concentrations of NFC and fat inversely reduced their impact on weight gain. These findings highlight the importance of a balanced nutrient composition for efficient organic waste conversion.

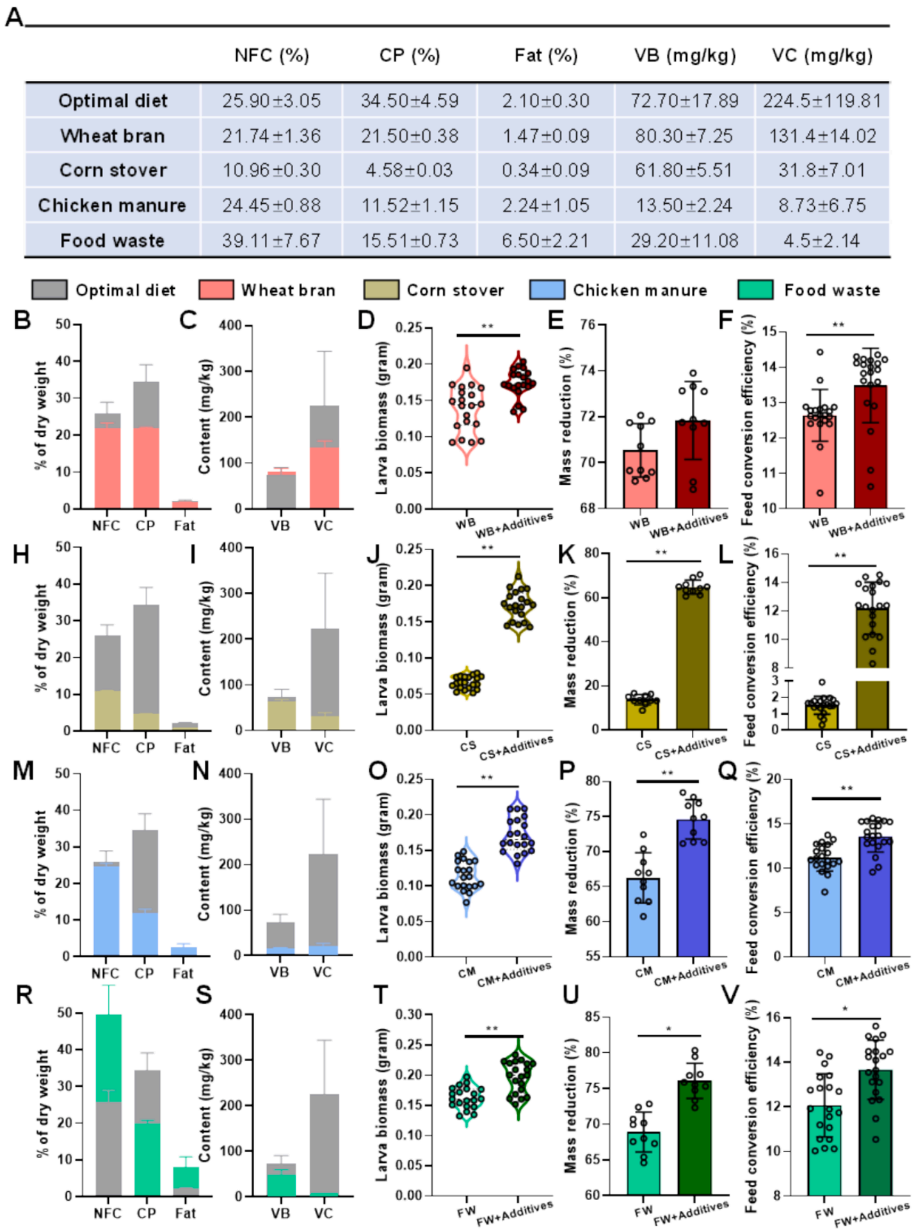
To validate the performance of the weight gain model, 30,000 diets with randomly assigned nutrient compositions (RD, Supplementary Table 1) were generated. The XGBoost model predicted larval biomass for these diets (Fig. 4F), and the top five diets with the highest predicted larval weights (RD11497, RD16781, RD3065, RD3728, and RD11337) were selected for feeding tests. Larvae fed these top five diets achieved significantly higher body weights than those of larvae fed other RDs (Fig. 4F). Although no significant difference was observed between the predicted and actual weight gain data (Fig. 4G), the actual observed weight gain was lower than the predictions made by the XGBoost model. This discrepancy could be attributed to several factors, including the physiological condition of the BSF larvae (e.g., larvae infested with pathogens may exhibit lower growth efficiency than healthy individuals), nutrient imbalances caused by experimental errors (such as inaccuracies in measuring nutrient factors), and environmental influences (e.g., air flow and texture of diets).

The predicted results of the 30,000 RDs identified the optimal nutrient model (RD11497), which consisted of 25.9 % NFC, 34.5 % CP, 2.1 % fat, 72.7 mg/kg B-group vitamins, and 224.5 mg/kg vitamin C. This composition closely resembles the response surface results. Collectively, the nutrient-based weight gain model constructed using XGBoost was shown to reliably replicate the growth of BSF larvae reared under laboratory conditions.

These findings demonstrate that protein content in the diet is the most critical factor influencing BSF larval growth, corroborating results from a previous study (Barragan-Fonseca et al. 2021). Protein content is also a major determinant of growth performance in other insect species, including caterpillars, honeybees, flies, and beetles (Silva-Soares et al. 2017; Kutz et al. 2019; Kotsou et al. 2021; Barraud et al. 2022; Hervet et al. 2023). Comparison between animal-source protein (casein) and plant-source protein (soy peptone) revealed no significant differences in BSF larval performance or waste conversion rates, indicating that BSF larvae can adapt to various protein sources. Proper mixing of organic wastes, particularly those with high protein content (including wheat bran in this study), can mitigate the nutrient imbalances of some BSF food sources (including corn stover), thereby improving waste recycling efficiency. This study tested nutrient intake in BSF larvae, including CP, carbohydrates, fat, and vitamins. The nutrient composition of new biowastes must be validated using the XGBoost model. It should be noted that nutrient elements such as fatty acids, amino acids, minerals, and other vitamins including vitamin D and vitamin E can also influence the end product and biowaste conversion rate of BSF. However, it is chanlleging to include a vast array of nutrients and measure their concentration combination in the diet in a single study. Only five major nutrient factors (the most ctritical macronutrients and vitamins commonly presented in the insect diet) were measured in this study. Meanwhile, these nutrient factors can be easily find in the organic wastes. For example, vegetable oils and animal fat in the food wastes; polysaccharides (starch and cellulose) and their microbial degradation products; and protein substances in chicken manure and wheat bran. These macronutrients embed in biowastes can be utilized by BSF larva. Additionally, the nutrient requirement of other nutrients and their impact on larval production will be further studied to update the database. Additionally, many other factors such as microbe community, environmental factors that influence insect development (Hatle et al. 2023) will also be considered to improve the performance of the model.

3.5. Appropriate mixed diet optimised by Machine learning model enhances the yield and organic waste bioconversion rate of BSF larvae

After determining the best nutrient model for BSF larvae, whether supplementing deficient nutrients in organic wastes could enhance composting efficiency was evaluated. Four types of organic waste were used: wheat bran, corn stover, chicken manure, and food waste (Fig. 5A). Nutrient deficiencies in each waste type were calculated



A

	NFC (%)	CP (%)	Fat (%)	VB (mg/kg)	VC (mg/kg)
Optimal diet	25.90±3.05	34.50±4.59	2.10±0.30	72.70±17.89	224.5±119.81
Wheat bran	21.74±1.36	21.50±0.38	1.47±0.09	80.30±7.25	131.4±14.02
Corn stover	10.96±0.30	4.58±0.03	0.34±0.09	61.80±5.51	31.8±7.01
Chicken manure	24.45±0.88	11.52±1.15	2.24±1.05	13.50±2.24	8.73±6.75
Food waste	39.11±7.67	15.51±0.73	6.50±2.21	29.20±11.08	4.5±2.14

Fig. 5. Impact of nutrient additives on BSF larvae performance when fed organic wastes, based on the optimal nutrition model. (A) Nutrient composition of the optimal BSF diet (Diet 11,497 with the highest larval biomass) compared to four common organic wastes used in BSF farming. Performance of BSF larvae fed on wheat bran (B–F), corn stover (G–K), chicken manure (L–P), and food waste (Q–U) is shown. (B, G, L, and Q) Comparison of macronutrient content (red for wheat bran, yellow for corn stover, blue for chicken manure, and green for food waste) with the optimal diet (grey). (C, H, M, and R) Comparison of vitamin content in each organic waste material. After adding feed additives to compensate for nutrient deficiencies in each organic waste, the following parameters were estimated: larval biomass (D, I, N, and S), mass reduction rate (E, J, O, and T), and feed conversion efficiency (F, K, P, and U). Variation analysis was performed using the Mann–Whitney *U* test. *, 0.01 < p < 0.05; **, p < 0.01. BSF, black soldier fly; NFC, non-fibre carbohydrates; CP, crude protein; VC, vitamin C; VB, vitamin B.

(Fig. 5B–C, G–H, L–M, and Q–R). Wheat bran and food waste contained higher CP levels than those of corn stover and chicken manure (wheat bran: 21.50 %; food waste: 15.51 %; corn stover: 4.58 %; chicken manure: 11.52 %). Food waste also had the highest NFC and fat content (39.11 % and 6.50 %, respectively) but was suboptimal in vitamins and CP. Wheat bran provided more vitamins than other wastes but lacked fat. Corn stover, commonly used in agriculture, was deficient in major nutrients required for BSF growth, including NFC, CP, and fat (Fig. 5A). After supplementing wheat bran with CP (casein) and vitamin C (Fig. 5B and C), larvae fed this balanced diet showed significantly higher average biomass (0.17 ± 0.02 g) compared with that of those fed unbalanced wheat bran (0.14 ± 0.03 g, Fig. 5D). Additionally, the mass reduction rate (Equation (9)) of waste and feed conversion efficiency (Equation (10)) of larvae improved significantly (Fig. 5E–F), indicating that nutrient supplementation enhanced the utilisation of wheat bran. Subsequently, corn stover with extra NFC (sucrose), CP, fat, and vitamin C (Fig. 5G and H) were supplemented before feeding it to BSF larvae. While larvae fed unbalanced corn stover exhibited very low biomass (0.06 ± 0.01 g), mass reduction (13.78 ± 2.19 %), and feed conversion rates (1.53 ± 0.53 %, Fig. 5I–K), all these indices markedly improved after supplementation. It is worth noting that the diet in this case was over supplemented, suggesting that BSF larvae may not have consumed the corn stover itself, and the observed improvements could be attributed to the additional nutrients provided. For chicken manure, CP, B-group vitamins, and vitamin C were added (Fig. 5L and M). Larvae fed this balanced diet achieved high biomass (Fig. 5N), along with improved mass reduction and feed conversion rates (Fig. 5O and P). Similarly, nutrient-balancing food waste with vitamin C and other additives (Fig. 5Q and R) increased larval weight and decomposition efficiency (Fig. 5S–U). These results demonstrate that adding feed supplements to organic wastes based on the optimised nutrient model can significantly improve BSF larval production and conversion rates. However, some supplements (including casein) used in this study are cost-prohibitive for large-scale BSF farming. Therefore, developing low-cost strategies to address this limitation is essential.

Compared to the nutrient requirements of BSF larvae (Fig. 5A), nutrient deficiencies vary significantly across commonly used organic wastes. For instance, the food waste in this study was deficient in vitamins, whereas wheat bran lacked fat. Addressing these deficiencies is crucial to improving larval weight gain. Mixing different organic wastes may be the most efficient and cost-effective strategy to fulfil dietary requirements. The model constructed in this study provides a quick and

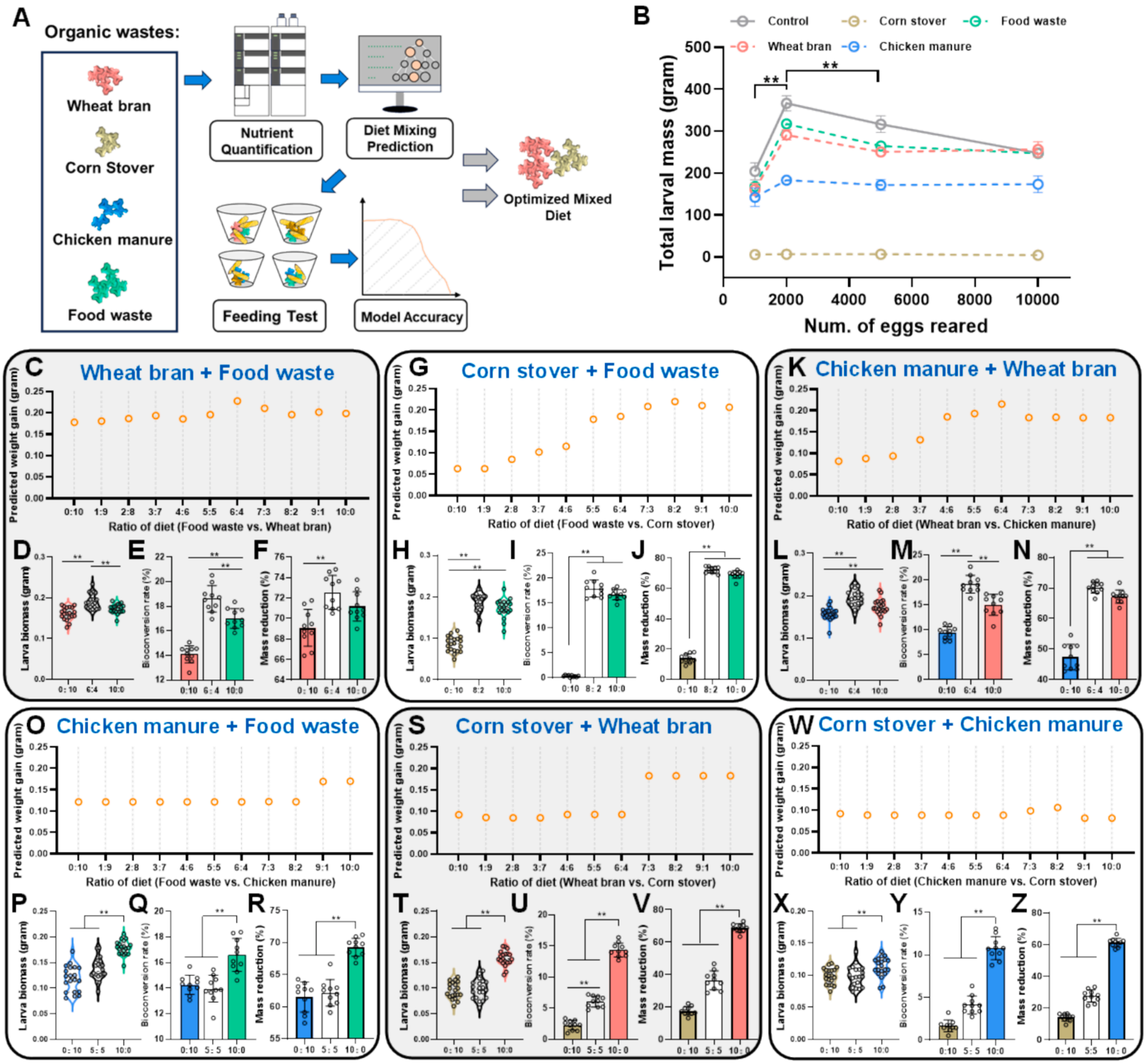


Fig. 6. Effect of biowaste mixture ratios on BSF larvae performance and biowaste bioconversion efficiency. (A) Experimental design for testing mixed biowaste diets at various ratios (0:10 to 10:0) to optimise BSF larval growth and biowaste conversion. Larval biomass was initially predicted by the XGBoost model and then confirmed by the feeding data. (B) The best feeding density was determined in the limited diet supplementation condition (1.5 kg diet). Different types of organic waste used for feeding BSF larvae are represented by lines in distinct colours. Variation analysis was calculated using the Mann–Whitney *U* test. **, p < 0.01. (C–F) Food waste + wheat bran mixtures: (C) Predicted larval biomass (XGBoost model), (D) Measured larval biomass, (E) Bioconversion rate, and (F) Mass reduction rate. (G–J) Food waste + corn stover mixtures: (G) Predicted larval biomass, (H) Measured larval biomass, (I) Bioconversion rate, and (J) Mass reduction rate. (K–N) Wheat bran + chicken manure mixtures: (K) Predicted larval biomass, (L) Measured larval biomass, (M) Bioconversion rate, and (N) Mass reduction rate. (O–R) Food waste + chicken manure mixtures: (O) Predicted larval biomass, (P) Measured larval biomass, (Q) Bioconversion rate, and (R) Mass reduction rate. (S–V) Wheat bran + corn stover mixtures: (S) Predicted larval biomass, (T) Measured larval biomass, (U) Bioconversion rate, and (V) Mass reduction rate. (W–Z) Chicken manure + corn stover mixtures: (W) Predicted larval biomass, (X) Measured larval biomass, (Y) Bioconversion rate, and (Z) Mass reduction rate. BSF, black soldier fly; XGBoost, extreme gradient boosting.

flexible approach to estimating total larval yield using various mixtures of organic wastes. To test this, two different waste materials in varying ratios (from 0:10 to 10:0) were combined and calculated larval biomass using the XGBoost weight gain model. The predictions were validated through feeding tests (Fig. 6A). An optimal feeding density was established to ensure that all dietary components were consumed. The results showed that feeding 2,000 BSF larvae with a 1.5 kg diet yielded the highest total biomass (Fig. 6B). This feeding density was used in subsequent mixed-diet experiments. The model predicted that combinations of food waste and wheat bran (FW + WB), food waste and corn stover (FW + CS), and wheat bran and chicken manure (WB + CM) would yield higher larval weights than those of diets consisting of a single material (Fig. 6C, G, and K). Other combinations, such as food waste and chicken manure (FW + CM), wheat bran and corn stover (WB + CS), and chicken manure and corn stover (CM + CS), showed no improvement upon mixing (Fig. 6O, S, and W). To confirm the predictions, BSF larvae were fed the best mixing ratios for each biowaste combination. For combinations that showed no improvement (FW + CM, WB + CS, and CM + CS), a 5:5 ratio was used. To ensure larvae did not preferentially consume one waste material over another (including only food waste in the FW + CS combination), a diet-limited feeding system was employed. The optimal feeding density of 2,000 larvae per 1.5 kg diet, as identified earlier, was maintained across all feeding tests.

In the FW + WB feeding group, the 6:4 ratio yielded the highest larval production (Fig. 6D). Additionally, larvae fed with this ratio exhibited the highest bioconversion rate (calculated by Equation (11)) of organic wastes (18.62 ± 1.02 %, Fig. 6E), which was significantly higher than that of larvae fed solely on food waste (17.02 ± 0.75 %) or wheat bran (14.09 ± 0.66 %). Compared with the 0:10 wheat bran diet (69.08 ± 1.70 %), the 6:4 mixed FW + WB diet demonstrated a 3.5 % increase in the mass reduction rate (72.54 ± 1.57 %, Fig. 6F). A similar increase was observed in the FW + CS diet, where a ratio of 8:2 achieved the highest efficiency among all mixing ratios (bioconversion rate: 17.98 ± 1.63 %; mass reduction rate: 72.36 ± 1.67 %, Fig. 6I and J). Remarkably, this diet-limited feeding test confirmed that corn stover could also be consumed by BSF larvae to provide nutrients that are deficient in food waste, resulting in an increase in larval weight with the 8:2 mixed diet (Fig. 6H). The improvements in larval production and decomposition rate were also validated in the WB + CM mixed group (Fig. 6K–N). Mixing WB and CM in a 6:4 ratio yielded a bioconversion rate of 19.18 ± 1.69 %, representing a 4.1 % improvement over wheat bran alone (15.03 ± 2.00 %) and a 9.7 % improvement over chicken manure alone (9.44 ± 1.17 %) (Fig. 6M). The mass reduction rate of the 6:4 WB + CM mix showed improvements of 22.78 % compared to CM alone and 4.15 % compared to WB alone (Fig. 6N).

In contrast, the FW + CM combination showed no positive effects after diet mixing (Fig. 6O–R), as also observed with the WB + CS mixing group (Fig. 6S–V). In these groups, a higher ratio of FW to WB led to higher bioconversion and mass reduction rates of organic waste. Similarly, the combination of chicken manure and corn stover, both CP-rich materials, showed no effect on weight gain or waste conversion efficiency (Fig. 6W–Z). Apart from the nutrient composition of the mixed diet, it should be noted that the texture of the diet can also influence the larval production. The corn straw or chicken manure diet showed a sticky texture with a low air permeability, which makes it difficult for BSF larvae to move within the feed (moisture content: 60 %). However, when wheat bran or food waste was added into these biowastes, the air permeability was increased, allowing BSF larvae to move and feed more easily. Therefore, the texture and air permeability of the feed should be considered initially in BSF farming process.

Collectively, feeding BSF larvae with combinations of organic wastes (with unbalanced nutrition) performed better (both weight gain, protein content and fat content in larvae, see Supplementary material) than feeding solely on nutrient-deficient biowastes. An appropriate mixed diet improved larval production and waste bioconversion efficiency. In this study, only four organic wastes that commonly used in the BSF farming industry were tested. The machine learning-based weight gain and diet nutrient model can efficiently predict the best mixing ratios of other organic wastes if their nutrient compositions are qualified. This enables optimisation of the nutrient quality of biowaste materials, thereby enhancing waste recycling and larval yield in BSF farming.

4. Conclusion

This study integrated machine learning algorithm (XGBoost) into BSF farming process, which helps estimating and optimising BSF diet nutrition. The optimal nutrient composition (25.9 % NFC, 34.5 % CP, 2.1 % fat, 72.7 mg/kg vitamin B mix, and 224.5 mg/kg vitamin C in the dry diet) was calculated and provided strategies to improve BSF waste conversion rates: precisely supplementing deficient nutrients in biowastes and mixing nutrient-biased biowastes with appropriate rates. These methods successfully elevated the larval feed efficiency and biowaste conversion rates (up to 9.7 %) while reducing feed material costs. These findings provide a robust framework for advancing BSF farming through cost-effective biowaste recycling and insect-based protein production.

CRediT authorship contribution statement

Shasha Feng: Writing – original draft, Methodology, Investigation, Data curation. **Hongyan Ma:** Methodology, Investigation, Data curation. **Chenxin Wu:** Validation, Investigation, Data curation. **Vimalanathan ArunPrasanna:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Xili Liang:** Validation, Investigation, Formal analysis, Data curation. **Dayu Zhang:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Conceptualization. **Bosheng Chen:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation of China (NSFC) (Grant No. 32100397) and Research Foundation of Zhejiang A&F University (Grant No. 2021LFR007) awarded to B.C.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.132254.

Data availability

Data will be made available on request.

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